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Policy paper

Critical Minerals technical annex

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Growth minerals

As outlined in the new UK Critical Minerals Strategy (<https://www.gov.uk/government/publications/uk-critical-minerals-strategy>), we have introduced a growth minerals list to better understand future demand of minerals vital to the UK economy, focusing and aligning both government and industry efforts towards achieving Vision 2035. Growth minerals are minerals deemed to be significant inputs for the Industrial Strategy growth-driving sectors (<https://www.gov.uk/government/collections/the-uks-modern-industrial-strategy-2025>), either due to their expected strong demand growth or because they cannot be substituted.

This allows government to not only support the current critical minerals list, but also to look ahead to the future and ensure that the UK is well-positioned to develop domestic capabilities and secure the supplies of minerals that will become increasingly important to the economy. For example, copper is not classed as critical but is fundamental for advanced manufacturing and the clean energy transition.

This list, tailored to the Industrial Strategy (<https://www.gov.uk/government/publications/industrial-strategy>), has allowed for the creation of demand estimates to signal to UK suppliers, downstream industries and international partners the scale of supply increase needed to fuel the Industrial Strategy growth-driving sectors (IS-8).

Identifying the list of growth minerals

The growth minerals list is based on a forward-looking assessment linked to the IS-8. The 82 candidate materials from the Critical Mineral Intelligence Centre's (CMIC's) 2024 Criticality Assessment (<https://nora.nerc.ac.uk/id/eprint/538471/>) were eligible for selection. End-use percentages of these minerals, as published in the criticality assessment, were complemented by stakeholder engagement (industry, government and academia) and wider desk-based research. These were then mapped to the IS-8 sectors.

This allowed for the identification of minerals that are key for both current and future industry, sectors and technologies. The resulting list of 23 growth minerals reflects those deemed to be the most significant for UK-based manufacturing related to the IS-8 growth-driving sectors.

This means that some critical minerals, such as iron and zinc, have not been classed as growth minerals. This is primarily driven by their predominant use in steelmaking, which is a foundational sector rather than part of the IS-8. Such critical minerals are still encompassed by the policy levers in this strategy, as well as being part of the upcoming Steel Strategy.

There is an overlap between the critical and growth minerals lists. This is driven by the high end-use percentage of IS-8 on minerals that are also assessed to be critical. Currently, the CMIC assesses minerals against 2 dimensions to determine criticality:

- global supply risk
- UK economic vulnerability

Some growth-driving sectors depend on minerals which are not currently assessed to be critical, this has resulted in new minerals such as copper and uranium to be classed as growth minerals.

The critical and growth mineral lists complement each other. Global supply risk and economic vulnerability is captured by the critical minerals list. Growth minerals identify the key future requirements for the UK economy. Together, both lists provide government with the ability to respond to global supply risk and adopt a forward-looking approach to securing critical minerals for the UK.

Demand signal methodology

For growth minerals, we have compiled signals of expected demand up to 2035 in line with the timescale of the vision for this strategy. The signals are tailored to UK manufacturing using raw and intermediate critical minerals, rather than the importing of end products for UK consumption, to estimate the needs of the IS-8.

Examples of products in scope of this demand analysis include:

- raw critical minerals, such as ores and concentrates
- chemicals, such as oxides, hydroxides and dioxides
- intermediate products, such as bars, rods, profiles and alloys

- waste and scrap, from which secondary supply can be salvaged and reintroduced to the production process via the circular economy

Products not in scope of this demand analysis include end products produced outside the UK, which are then imported to the UK with embedded critical minerals. For example, completed electric vehicle (EV) batteries and finished consumer electronic products.

Annex 2 sets out UK-specific estimates of growth mineral demand through to 2035. These demand signals are a first step in understanding UK demand, upon which future developments can be made. Due to sensitivities, demand estimates for the Defence sector have been excluded, although some of this demand is likely captured by Advanced Materials within the Advanced Manufacturing sector.

The methodology for creating demand signals can be found in Annex 4 on a mineral-by-mineral basis. A summary of the hierarchy of evidence that was used to construct these demand signals is as follows.

1. UK-specific estimates for 2025 to 2035 growth mineral demand from official sources

We looked at official sources, such as the British Geological Survey's Critical Mineral Intelligence Centre (CMIC), or the Advanced Propulsion Centre (APC) (<https://www.apcuk.co.uk/>). Some of this data, such as CMIC's foresight studies into various advanced manufacturing and clean energy technologies, considers various growth rate scenarios based on the National Grid's Future Energy Scenarios^[footnote 1]. The 3 scenarios chosen for this analysis are: System Transformation, Consumer Transformation, and a midpoint between the 2 to capture the pathways to Net Zero. Additionally, other studies on aerospace alloys (CMIC) and semiconductors (UK government) provide various growth-rate scenarios which have been incorporated into these demand signals.

2. End-use percentages from CMIC's 2024 Criticality Assessment

End-use percentages for growth minerals, taken from CMIC's 2024 Criticality Assessment, were applied to UK imports to attribute these to IS-8 growth sectors. While this data lacks a variety of scenarios and dynamic growth rates, it does fill in data gaps to provide a more representative picture of the scale of UK growth mineral needs to support the IS-8 sectors.

3. Paid-for market data sources

Paid-for market data sources, such as Benchmark Mineral Intelligence (<https://www.benchmarkminerals.com/>), and trade associations such as the International Tin Association (<https://www.internationaltin.org/>), were also consulted for insight. These data sources provide dynamic growth rates or a range of demand scenarios, or both. In line with transparency guidelines, data that is not publicly available has not been used in the demand signal calculations, but was used for comparison against which to review Department for Business and Trade (DBT) demand signals.

Future developments

Data related to critical minerals, especially for current and future demand, is a developing space. These demand signals have been constructed in consultation with industry stakeholders, other government departments, and CMIC, using the best data available at the time of publication. This work has been peer reviewed by government economists and CMIC.

DBT will continue to develop this demand forecasting analysis related to growth minerals. This includes ongoing cross-government work as part of the Global Clean Power Alliance (GCPA) to improve data relevant for clean energy supply chains via international collaboration^[footnote 2]. DBT is also working with the Office for National Statistics (ONS) to better define the UK critical mineral sector under a refresh of the Classification of Products by Activity (CPA) codes, as well as a wider consultation of Standard Industrial Classification (SIC) codes. Annex 6 contains an early definition of the critical mineral sector under the current SIC07 definitions.

Annex 1

Complete list of UK critical minerals

The complete list of UK critical minerals, sourced from the Critical Mineral Intelligence Centre's 2024 Criticality Assessment, is as follows:

- aluminium
- antimony
- bismuth
- borates
- cobalt
- gallium
- germanium
- hafnium
- helium
- indium
- iridium
- iron
- lithium
- magnesite
- magnesium
- manganese
- natural graphite
- nickel
- niobium
- phosphorus
- platinum
- Rare Earth Elements (REEs)
- rhenium
- rhodium
- ruthenium
- silicon
- sodium
- tantalum
- tellurium
- tin
- titanium
- tungsten
- vanadium

- zinc

Complete list of UK growth minerals

The complete list of UK growth minerals is as follows:

- aluminium
- antimony
- beryllium
- chromium
- cobalt
- copper
- gallium
- germanium
- graphite (natural and synthetic)
- hafnium
- helium
- lithium
- manganese
- nickel
- Platinum Group Metals (PGMs) – which include iridium, platinum, rhodium and ruthenium
- Rare Earth Elements (REEs)
- rhenium
- silicon
- tantalum
- tin
- titanium
- tungsten
- uranium

Annex 2

Table 1: Industrial Strategy growth sector cumulative demand estimates (tonnes) for 2025 growth mineral list

Mineral	2024	2027 midpoint estimate	2030 midpoint estimate	2035 midpoint estimate
Aluminium	578,300	1,875,000	3,966,000	8,003,000
Antimony	428	1,284	2,568	4,707
Beryllium	8	24	47	87
Chromium	13,900	45,030	90,740	168,700
Cobalt	6,089	34,010	76,610	163,000
Copper	178,400	922,200	1,953,000	3,619,000
Gallium	1	4	9	21
Germanium	4	13	28	58
Graphite (natural and synthetic)	9,070	28,610	137,800	449,100
Hafnium	18	62	138	304
Helium	2,196	7,327	17,480	44,750
Lithium (LCE)	2,525	28,680	113,400	339,200
Manganese	18,840	56,920	118,800	227,500
Nickel	50,430	182,700	416,100	867,200
Platinum-group metals (PGMs)	2,710	7,643	15,290	28,040
REEs	1,161	7,788	18,020	37,940
Rhenium	1	4	8	18

Mineral	2024	2027 midpoint estimate	2030 midpoint estimate	2035 midpoint estimate
Silicon	57,500	210,800	425,000	785,800
Tantalum	65	421	867	1,682
Tin	4,768	8,927	18,370	35,580
Titanium	49,480	177,100	360,700	684,900
Tungsten	925	3,185	6,462	12,190
Uranium	710	2,149	5,425	10,200

Table notes

- Platinum-group metals (PGMs) include iridium, platinum, rhodium and ruthenium.

Annex 3

Figure 1: Expected cumulative aluminium demand by Industrial Strategy growth sector, tonnes, midpoint estimates

Change to table view

Aluminium	Electric Vehicles	Other Advanced Manufacturing	Clean Energy	Life Sciences	Digital and Technology
2027	78,200	1,735,000	62,060	0	0
2030	402,700	3,470,000	123,600	0	0
2035	1,451,000	6,361,000	191,200	0	0

Figure 2: Expected cumulative copper demand by Industrial Strategy growth sector, tonnes, midpoint estimates

[Change to table view](#)

Copper	Electric Vehicles	Other Advanced Manufacturing	Clean Energy	Life Sciences	Digital and Technology
2027	31,910	232,600	522,700	0	105,000
2030	106,900	465,100	1,171,000	0	209,900
2035	325,700	852,700	2,056,000	0	384,800

Figure 3: Expected cumulative lithium demand by Industrial Strategy growth sector, tonnes, midpoint estimates

[Change to table view](#)

Lithium	Electric Vehicles	Other Advanced Manufacturing	Clean Energy	Life Sciences	Digital and Technology
2027	21,000	107	0	0	7,574
2030	98,000	213	0	0	15,150
2035	311,000	391	0	0	27,770

Figure 4: Expected cumulative nickel demand by Industrial Strategy growth sector, tonnes, midpoint estimates

[Change to table view](#)

Nickel	Electric Vehicles	Other Advanced Manufacturing	Clean Energy	Life Sciences	Digital and Technology
2027	18,000	157,000	644	0	7,037
2030	84,000	316,900	1,170	0	14,070
2035	248,000	591,600	1,822	0	25,800

Figure 5: Expected cumulative Rare Earth Elements demand by Industrial Strategy growth sector, tonnes, midpoint estimates

[Change to table view](#)

Rare Earth Elements	Electric Vehicles	Other Advanced Manufacturing	Clean Energy	Life Sciences	Digital and Technology
2027	1,205	3,930	2,652	0	0
2030	3,345	7,861	6,814	0	0
2035	8,556	14,410	14,980	0	0

Figure 6: Expected cumulative silicon demand by Industrial Strategy growth sector, tonnes, midpoint estimates

Change to table view

Silicon	Electric Vehicles	Other Advanced Manufacturing	Clean Energy	Life Sciences	Digital and Technology
2027	349	163,900	37,960	0	8,626
2030	1,623	327,800	78,310	0	17,250
2035	5,445	601,000	147,700	0	31,630

Annex 4

Table 2: Summary of calculation methodology

Mineral	Calculation methodology
Aluminium	Electric vehicles (EV): UK EV production estimates for 2025 to 2035 from the Advanced Propulsion Centre (APC) were combined with aluminium content per vehicle data from European Aluminium/Ducker^[footnote 3] to calculate battery and vehicle body material requirements. Cumulative total for 2035: 1,451,000 tonnes. Advanced Manufacturing (AM): 3-year average of UK imports data from HM Revenue and Customs (HMRC) using HS codes in Annex 5, plus UK primary and secondary aluminium production from the <u>British Geological Survey (BGS) UK Mineral Yearbook</u>, assigned to sectors using CMIC end-use percentages as an estimate for current

Mineral	Calculation methodology
	consumption. Cumulative total for 2035: 6,361,000 tonnes. Clean Energy (CE): estimates taken from CMIC foresight studies on heat pumps and electricity grid decarbonisation. Cumulative total for 2035: 138,000 to 244,300 tonnes.
Antimony	AM, Digital and Technologies (DT): 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages as an estimate for current consumption. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative total for 2035: 2,354 tonnes per sector.
Beryllium	AM, DT: 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages as an estimate for current consumption. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative totals for 2035: 51 tonnes (AM) and 36 tonnes (DT).
Chromium	AM: 3-year average of UK import data based on HS codes in CMIC's aerospace alloys report, with applied growth rates of 0%, 7% and 9% cited in that report, the latter 2 being cited in the CMIC report. Cumulative total for 2035: 9,662 to 16,810 tonnes. AM: for non-aerospace uses, 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages as an estimate for current consumption. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative total for 2035: 153,400 tonnes. CE: taken from CMIC foresight study into nuclear energy. Cumulative total for 2035: 401 to 455 tonnes.
Cobalt	EV: estimates of UK cobalt demand for 2025 to 2035 taken from the <u>Advanced Propulsion Centre (Quarter 2 2025 report)</u>. Cumulative total for 2035: 34,650 tonnes. AM: 3-year average of UK import data based on HS codes in CMIC's aerospace alloys report, with applied growth rates of 0%, 7% and 9% cited in that report, the latter 2

Mineral	Calculation methodology
	being cited in the CMIC report. Cumulative total for 2035: 44,680 to 77,750 tonnes. AM, DT: 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages as an estimate for current consumption. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative total for 2035: 59,700 tonnes.
Copper	EV: copper foil for batteries sourced from Advanced Propulsion Centre (Quarter 2 2025 report). Cumulative total for 2035: 111,500 tonnes. Traction motors estimate from CMIC foresight study. Cumulative total for 2035: 174,200 to 254,300 tonnes. AM, DT, CE: 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages an estimate for current consumption. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative total for 2035: 1,996,000 tonnes. CE: taken from CMIC foresight studies into electricity grid decarbonisation, wind turbines, photovoltaics and heat pumps. Cumulative total for 2035: 895,900 tonnes to 1,699,000 tonnes.
Gallium	CE, DT: 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages as an estimate for current consumption. For DT, a range of semiconductor industry compound annual growth rate (CAGR) scenarios from a UK government study^[footnote 4] were applied to these imports. Constant rate of demand assumed for CE. Cumulative total for 2035: 18 to 24 tonnes.
Germanium	AM, DT: 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages as an estimate for current consumption. For DT, a range of semiconductor industry CAGR scenarios from a UK government study were applied to these imports. Cumulative total for 2035: 52 to 65 tonnes.
Graphite (natural and	EV: estimates of UK graphite demand, both natural and synthetic, for 2025 to 2035 taken from APC Quarter 2 2025

Mineral	Calculation methodology
synthetic)	report. Cumulative total for 2035: 447,500 tonnes. CE: estimates from CMIC's foresight study into fuel cell technology. Cumulative total for 2035: 483 to 2,639 tonnes.
Hafnium	AM: 3-year average of UK import data based on HS codes in CMIC's aerospace alloys report, with applied growth rates of 0%, 7% and 9% cited in that report, the latter 2 being cited in the CMIC report. Cumulative total for 2035: 198 to 344 tonnes.
Helium	Imports of helium were used and allocated to industries based on end-use data from Global Helium via Statista. Growth rates were applied to these, taking account of UK government's pledge to double the number of NHS MRI scanners, as well as the expected CAGR of the semiconductor industry where helium is used.
Lithium	EV: estimates of UK lithium demand for 2025 to 2035 taken from APC Quarter 2 2025 report. Cumulative total for 2035: 311,000 tonnes of lithium carbonate equivalent (LCE). AM, DT: 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages an estimate for current consumption. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative total for 2035: 28,160 tonnes.
Manganese	EV: estimates of UK manganese demand for 2025 to 2035 taken from APC Quarter 2 2025 report. Cumulative total for 2035: 30,900 tonnes. AM: 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative demand for 2035: 196,600 tonnes. CE: taken from CMIC foresight study into nuclear energy. Cumulative total for 2035: 8 to 9 tonnes.
Nickel	EV: estimates of UK nickel demand for 2025 to 2035 taken from APC Quarter 2 2025 report. Cumulative total for 2035: 248,000 tonnes.

Mineral	Calculation methodology
	AM: 3-year average of UK import data based on HS codes in CMIC's aerospace alloys report, with applied growth rates of 0%, 7% and 9% cited in that report, the latter 2 being cited in the CMIC report. Cumulative total for 2035: 40,840 to 71,060 tonnes. For non-aerospace uses, 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative total for 2035: 528,900 tonnes. CE: taken from CMIC foresight studies into nuclear energy and green hydrogen. Cumulative total for 2035: 1,361 to 2,284 tonnes. DT: 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages an estimate for current consumption. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative total for 2035: 25,800 tonnes.
Platinum Group Metals (PGMs)	AM: 3-year average of UK import data based on HS codes in CMIC's aerospace alloys report, with applied growth rates of 0%, 7% and 9% cited in that report, the latter 2 being cited in the CMIC report. Cumulative total for 2035: 25 to 43 tonnes. CE: taken from CMIC foresight study into green hydrogen. Cumulative total for 2035: 0.5 to 1.3 tonnes. AM, DT, Life Sciences (LS): 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages as an estimate for current consumption. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative total for 2035: 28,000 tonnes.
Rare Earth Elements (REEs)	EV: data from CMIC foresight study on traction motor technology. While this is not solely for EV uses, it is expected this will be the primary use of this technology. Cumulative total for 2035: 6,825 to 10,290 tonnes.

Mineral	Calculation methodology
	AM: 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages as an estimate for current consumption. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative total for 2035: 14,410 tonnes. CE: data from CMIC foresight studies into wind turbines, nuclear energy, heat pumps and green hydrogen. Cumulative totals for 2035: 12,940 to 17,010 tonnes.
Rhenium	AM: 3-year average of UK import data based on HS codes in CMIC's aerospace alloys report, with applied growth rates of 0%, 7% and 9% cited in that report, the latter 2 being cited in the CMIC report. Cumulative total for 2035: 12 to 21 tonnes.
Silicon	EV: based on data from CMIC study on traction motor technology. Cumulative total for 2035: 5,445 tonnes. AM, CE, DT: 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages as an estimate for current consumption. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative total for 2035: 727,500 tonnes. CE: taken from CMIC foresight study into nuclear energy and photovoltaics. Cumulative total for 2035: 42,240 to 63,470 tonnes.
Tantalum	EV, AM, LS, DT: capacitor consumption based on estimates from Mordor Intelligence. AM: 3-year average of UK import data based on HS codes in CMIC's aerospace alloys report, with applied growth rates of 0%, 7% and 9% cited in that report, the latter 2 being cited in the CMIC report. Cumulative total for 2035: 355 to 617 tonnes.
Tin	AM, CE, LS, DT: DBT calculations based on data supplied by the <u>International Tin Association</u>.

Mineral	Calculation methodology
Titanium	AM: 3-year average of UK import data based on HS codes in CMIC's aerospace alloys report, with applied growth rates of 0%, 7% and 9% cited in that report, the latter 2 being cited in the CMIC report. Cumulative total for 2035: 91,240 to 158,800 tonnes. AM, LS: 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages as an estimate for current consumption. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative total for 2035: 544,300 tonnes.
Tungsten	AM: 3-year average of UK import data based on HS codes in CMIC's aerospace alloys report, with applied growth rates of 0%, 7% and 9% cited in that report, the latter 2 being cited in the CMIC report. Cumulative total for 2035: 1,307 to 2,274 tonnes. For non-aerospace uses, 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative total for 2035: 10,180 tonnes.
Uranium	CE: taken from CMIC foresight study into nuclear energy. Cumulative total for 2035: 9,268 to 9,511 tonnes. AM, LS: 3-year average of UK imports (HMRC) of HS codes in Annex 5, assigned to sectors using CMIC end-use percentages as an estimate for current consumption. In the absence of data on growth rates, a constant level of demand each year was assumed to calculate a cumulative figure. Cumulative total for 2035: 809 tonnes.

Table notes

- The 2035 cumulative totals are rounded and may not add up to 100% of the values in Table 1.

Annex 5

Table 3: Harmonised System (HS) code definition of critical and growth minerals

Mineral	HS code
Aluminium	26060000, 26204000, 28182000, 28183000, 28261200, 28263000, 28273200, 28332200, 76011000, 76011010, 76011090, 76012030, 76012040, 76012080, 76020011, 76020019, 76020090, 76031000, 76032000, 76041010, 76041090, 76042100, 76042910, 76042990, 76051100, 76051900, 76052100, 76052900, 76061130, 76061150, 76061191, 76061193, 76061199, 76061230, 76061250, 76061292, 76061293, 76061299, 76069100, 76069200, 76072091, 76090000, 76169910, 76169990
Antimony	26171000, 28258000, 81101000, 81102000, 81109000
Beryllium	81121200, 81121300, 81121900
Bismuth	81061010, 81061090, 81069010, 81069090
Borates	25280000, 28045010, 28100010, 28100090
Chromium	26100000, 28191000, 28199010, 28199090, 72024110, 72024190, 72024910, 72024950, 72024990, 72025000, 81122110, 81122190, 81122200, 81122900
Cobalt	26050000, 28220000, 28273930, 28332930, 81052000, 81053000, 81059000
Copper	26030000, 26203000, 28255000, 28274100, 28332500, 28342940, 74010000, 74020000, 74031100, 74031200, 74031300, 74031900, 74032100, 74032200, 74032900, 74040010, 74040091, 74040099, 74050000, 74061000, 74062000, 74071000, 74072110, 74072190, 74072900, 74081100, 74081910, 74081990, 74082100, 74082200, 74082900, 74091100, 74091900, 74092100, 74092900, 74093100, 74093900, 74094000, 74099000, 74101100, 74101200, 74102100, 74102200, 74111010, 74111090, 74112110, 74112190, 74112200, 74112900, 74121000, 74122000, 74130000, 74192000, 74198010, 74198030, 74198090, 85441110, 85441190, 85446010
Gallium	81129289

Mineral	HS code
Germanium	81129295, 81129940
Graphite	25041000, 25049000, 38011000, 38012010, 38012090, 38019000, 68151900, 85451900, 85459090
Hafnium	81123100, 81123900
Helium	28042910, 28454000
Indium	81129281
Iridium	71104100, 71104900
Iron	25020000, 26011100, 26011200, 26012000, 26180000, 26190020, 28211000, 28273920, 72011011, 72011019, 72011030, 72011090, 72012000, 72015010, 72015090, 72041000, 72043000, 72044110, 72044191, 72044199, 72044910, 72044930, 72044990, 72045000, 72051000, 72052900, 72061000, 72069000, 73269098
Lithium	25309040, 28252000, 28369100, 38249975
Magnesite	25191000, 25199010, 25199090
Magnesium	25302000, 28161000, 28273100, 28332100, 81041100, 81041900, 81042000, 81043000, 81049000
Manganese	26020000, 26190020, 28201000, 28209010, 28209090, 72021120, 72021900, 72023000, 81110011, 81110019, 81110090
Nickel	26040000, 26209910, 28254000, 28273500, 28332400, 72026000, 75011000, 75012000, 75021000, 75022000, 75030010, 75030090, 75040000, 75051100, 75051200, 75052100, 75052200, 75061000, 75062000, 75071100, 75071200, 75072000, 75081000, 75089000
Niobium	26209920, 72029300, 81129240, 81129950
Platinum	71101100, 71101910, 71101980, 71129200, 71151000
Phosphates	25101000, 25102000, 28047000, 28047010, 28047090, 28092000, 28139010, 28351000, 28352200, 28352400, 28352500, 28352600, 28352910, 28352930, 28352990, 28353100, 28353900, 31031100, 31031900

Mineral	HS code
Rare Earth Elements (REEs)	28053010, 28053020, 28053021, 28053029, 28053030, 28053031, 28053039, 28053040, 28053080, 28461000, 28469010, 28469020, 28469030, 28469040, 28469050, 28469060, 28469070, 28469090, 36069010, 85051110
Rhenium	81124110, 81124190, 81124900
Rhodium	71103100, 71103900
Ruthenium	71104100, 71104900
Silicon	28046100, 28046900, 28112200, 28399000, 28492000, 38180010, 39100000, 72022100, 72022910, 72022990
Sodium	28051100, 28151100
Tantalum	81032000, 81033000, 81039910, 81039990, 85322100
Tellurium	28045090
Tin	26090000, 26209940, 28273910, 80011000, 80012000, 80020000, 80030000, 80070010, 80070080
Titanium	26140000, 26209960, 28230000, 28332930, 72029100, 81082000, 81083000, 81089030, 81089050, 81089060, 81089090
Tungsten	26110000, 28259040, 28418000, 28499030, 72028000, 81011000, 81019400, 81019600, 81019700, 81019910, 81019990
Uranium	26121090, 28441010, 28441030, 28442059, 28443011, 28443019, 28443091, 28444210, 28444310
Vanadium	26190095, 28253000, 72029200, 81129291
Zinc	26080000, 26201100, 26201900, 28170000, 79012000, 79020000, 79031000, 79039000, 79070000

Annex 6

Table 4: DBT UK critical mineral sector definition, including the 4 non-critical growth minerals

UK industry (SIC 2007)	Total gross value added (GVA) (£ millions)	Total employment	Proportion	Weighting	Weighting GVA million
0710: Mining of iron ores	0	0	100	1	0
0721: Mining of uranium and thorium ores	0	0	50	0.5	0
0729: Mining of other non- ferrous metal ores	Suppressed	0	100	1	No available
1920: Petroleum refining (petroleum coke, of which the UK is a key global producer, is a key input for synthetic graphite)	2,986	9,000	9	0.09	28

UK industry (SIC 2007)	Total gross value added (GVA) (£ millions)	Total employment	Proportion	Weighting	Weighting GVA mil
2410: Manufacture of basic iron and steel and of ferro- alloys	Suppressed	25,000	100	1	No ava
2441: Precious metals production	131	400	33	0.33	43.
2442: Aluminium production	Suppressed	4,500	90	0.9	No ava
2443: Lead, zinc and tin production	108	1,000	66	0.66	72
2444: Copper production	95	1,500	100	1	95
2445: Other non-ferrous metal production	355	3,000	100	1	35.
2446: Processing of nuclear fuel	Suppressed	14,000	100	1	No ava
3812: Collection of hazardous waste	108	1,750	50	0.5	54
3832: Recovery of	2,674	21,000	33	0.33	89.

UK industry (SIC 2007)	Total gross value added (GVA) (£ millions)	Total employment	Proportion	Weighting	We GV mil
sorted materials					
Total	6,457	81,150	Not applicable	Not applicable	1,7

Table notes

- Weightings have been applied to better represent the contribution of critical or growth minerals to these industries. Proportions and weightings have been calculated using a combination of sectoral knowledge and Standard Industrial Classification (SIC) 2007 methodology notes from the Office for National Statistics.
- Some data has been suppressed to preserve confidentiality.

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